to Markowitz (1952) to solve this trader's dilemma by balancing the trade-off between market impact cost (*MI*) and timing risk (*TR*). Mathematically, these optimal trading strategies are computed as follows:

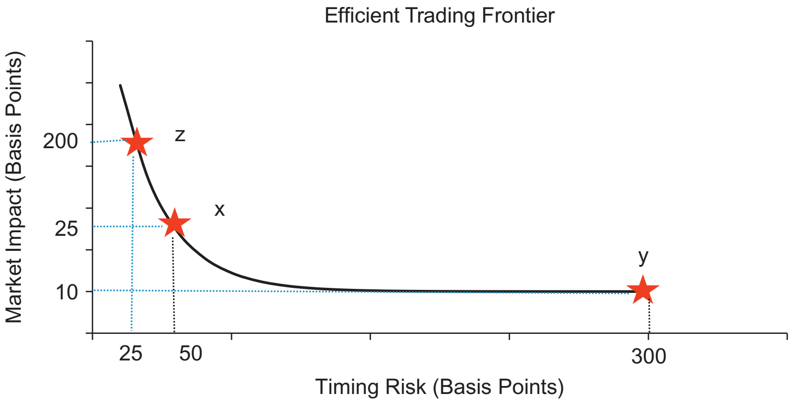
$$Min \quad MI(x) + \lambda \cdot TR(x) \tag{10.4}$$

where x denotes the optimal trading schedule (e.g., how shares are to be transacted over the trading horizon) and λ denotes the investor's level of risk aversion. The appropriate formulation of the market impact function and portfolio optimization techniques have been the focus of earlier chapters.

If we solve Equation 10.4 for all values of $\lambda \geq 0$ we obtain the set of all optimal strategies. When plotted, these strategies constitute the Almgren and Chriss efficient trading frontier (ETF). This is illustrated in Figure 10.3. In this example, we have highlighted three different strategies. Strategy x denotes a moderately paced trade schedule with market impact 25 bp and timing risk 50 bp, strategy z is an aggressive strategy with high market impact 200 bp but low timing risk 25 bp, and strategy y is a passive strategy, e.g., VWAP, with low market impact 10 bp but much higher timing risk 300 bp. The efficient trading frontier is illustrated in Figure 10.3.

What is the appropriate optimal strategy to use?

There has been quite a bit of research and industry debate focusing on how to determine the appropriate optimal strategy. Bertsimas and Lo (1998) propose minimizing the combination of market impact and price appreciation (price drift) without regards to corresponding trading risk.



■ **Figure 10.3** Efficient Trading Frontier.